

Reinforcement Learning For Trading

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CSE 5100: Deep Reinforcement Learning

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1 Project Description

In financial markets, there are many different trading strategies coming from different fields, including technical analysis, fundamental analysis, machine learning, and so on. Our group wants to test out Reinforcement Learning's performance in trading.

There are mainly four motivations for our team:

- The financial trading scenario naturally fits into the reinforcement learning framework. The profit and loss for each step is a clear and natural candidate for determining the step rewards.
- Each market opportunity will gradually disappear (alpha decay) as more competitors uncover the same strategies. It is less likely to find real alpha from traditional approaches. Therefore, we want to examine if Reinforcement Learning, a relatively new method, could uncover some useful results.
- Unlike statistical methods and supervised learning's two step's approach that first predicts price movements and then separately decides trade positions, Reinforcement learning bypasses the prediction step entirely and directly optimizes trading decisions, potentially uncovering strategies that are difficult to derive from conventional methods.
- As a loose analogy, AlphaGo showed that RL can discover strategies in a competition setting that are hard for human to discover. While financial markets are far more complex, we are curious whether RL can similarly find trading patterns that human might miss, since financial markets can essentially be viewed as a multiplayer competition.

In this project, we aim to mainly focus on revising and improving the reinforcement learning strategies from the article "Deep Reinforcement Learning for Trading" by Zhang et al. [2019] Since the article was published in 2020 and gives backtest results from 2011–2019, our group wants to first replicate the procedure and then test the algorithms after 2019 to see how performance changes. As mentioned above, one issue in financial markets is alpha decay — strategies tend to lose their edge over time as more participants discover and exploit them. Since the original paper reports strong results from 2011–2019, a natural question is whether those results still hold after the paper was published and the methods became publicly known. By testing the algorithms on data from 2019–2025, we can better evaluate how robust these RL strategies truly are.

Then we want to make adjustments to the reward definition, asset classes, feature set, and so on from the paper to see if we can improve the algorithms' performance. Eventually, we may also implement some new trading strategies using RL. For example, pairs trading could be a good candidate for RL. In pairs trading, we assets whose price spreads have historically been mean reverting. Compared to trading a single asset where the price can drift in any direction, a mean-reverting spread gives the RL agent a more predictable and stable state space to learn from, which could make it easier for the agent to discover effective trading rules.

2 Existing Literature

There are many articles on using RL in trading. The article we mainly build on is "Deep Reinforcement Learning for Trading" by Zhang et al. [2019]. In addition, we also find some papers on using RL in pairs trading, such as "Deep Reinforcement Learning Pairs Trading with a Double Deep Q-Network" by Brim [2020]. We will test it on different asset classes and over a different time period. In addition, we will also try implementing strategies other than pairs trading from this article.

3 Deliverables

1. **Project Report:** A written report with introduction, methodology, results, and analysis.
2. **Presentations:** A presentation with brief introduction and methodology we used and mainly focus on our key findings of strategy performance across different time periods and asset classes.
3. **Code** We will provide the code repository with all implementations.

4 Required Resources

This project requires moderate amount of computational resources. More effort will be devoted to experimentation. The main resources needed are historical financial data, compute for training and evaluation, and time for debugging and analysis. Specifically, WashU’s Research Infrastructure Services will be used, which provides both compute (GPU and CPU) and storage resources. Model’s training time is not to be specified here since it would be highly dependent on the evaluation results, but the upper limit of the training time of a single model for this project will be bounded below 24 hours. Overall, we expect the total GPU hours used for this project to be within 40 hours.

We will use historical market data for the selected assets, including the original evaluation period used by prior work and a newer post-2019 test period. Data storage and preprocessing can be handled on a personal computer, which is more convenient than the compute nodes. For debugging, and smaller experiments, we will use our local machines as well. For larger training runs, hyperparameter sweeps, and repeated evaluations across time periods and reward formulations, the cluster will be used

5 Tentative Plan

This project is scheduled to last for one month. The detailed weekly plan is as follows:

- Week 1** Read the reference papers carefully, finalize the exact scope of the project. Set up the historical data pipeline and simulated trading environment. Define the evaluation metrics based on reference papers
- Week 2** Implement and reproduce the reinforcement-learning trading method mentioned in the primary reference as baseline. Run initial experiments to confirm that the agent and environment behave reasonably.
- Week 3** Evaluate the reproduced baseline on post-2019 data and compare it with simple non-DRL benchmark strategies. Modify and test the reward function if time permits.
- Week 4** Analyze the results, create final figures and tables, and prepare the report and presentation/poster. Clean the codebase and document the experimental setup, and upload to GitHub.

References

- Andrew Brim. Deep reinforcement learning pairs trading with a double deep q-network. In *2020 10th Annual Computing and Communication Workshop and Conference (CCWC)*, pages 0222–0227, 2020. doi: 10.1109/CCWC47524.2020.9031159.
- Zihao Zhang, Stefan Zohren, and Stephen Roberts. Deep reinforcement learning for trading, 2019. URL <https://arxiv.org/abs/1911.10107>.